HW7 homework debrief.

We ended up discussing more of problems from the final exam packet, rather than from HW7 itself.

If A and B commute, must A^2 and B^2 commute?

Yes, since $A^2B^2 = AABB = A(BA)B = (BA)AB =$

$$BA(BA) = B(BA)A = B^2A^2.$$

What about the converse? If A^2 and B^2 commute, must A and B?

Not necessarily.

For instance, any pair of reflections in $\mathbb{R}^2$ across lines that are not perpendicular do not commute. (the two products are rotations by the same amount but in opposite directions.) But their squares commute, since each square is just the identity.

Sources of commuting matrices

An $n \times n$ identity matrix commute with any other $n \times n$ matrix.

Same for the $n \times n$ zero matrix, or kI for any scalar k.

Rotations about the same axis commute.

Any square matrix commutes with itself, powers of itself, and polynomials of itself.